Advanced Financial Modeling, Module 11: Equity vs. Debt vs. Convertible Bond Analysis from a Blank Sheet (Netflix) (1-Week Case Study)

You'll complete a case study based on Netflix in this module, which has several components.

In the first part, you'll start from a blank Excel sheet and build a 3-statement model and credit/refinancing analysis for the company to recommend the best way to treat its upcoming $4 billion debt maturities.

In the second part, you'll analyze one of the company's specific bond issuances and calculate metrics such as the Yield to Maturity (YTM), Duration, and Convexity, and recommend whether the company should refinance that issuance right now.

Finally, you'll conclude by analyzing a potential convertible bond issuance for the company, including a valuation, an estimate of its blended cost, and dilution estimates at different share prices if Netflix sets up a "capped call" transaction.

AFM-11-01: Netflix Equity vs. Debt vs. Convertible Bond Case Study Overview and Strategy (15:00)

In this lesson, you'll get an introduction to this case study and learn about the decision-making process for debt and equity financings and the recommended model completion order.

Table of Contents:

- **2:24:** Part 1: How to Interpret the Case Document
- **5:04:** Part 2: How to Make a Financing Decision
- **7:35:** Part 3: Recommended Model Order and Complexity
- **13:23:** Recap and Summary

Quiz Questions:

[No quiz questions.]

AFM-11-02: Inputting and Simplifying the Financial Statements (29:31)

You'll learn how to link in, simplify, and format Netflix's historical financial statements in this lesson based on the Excel or CSV-formatted financials on the company's website.

Table of Contents:

- **2:34:** Income Statement

- **12:23:** Balance Sheet
- **18:45:** Cash Flow Statement
- **27:53:** Recap and Summary

Quiz Questions:

12:22 What is the MOST EFFICIENT way to input a company's historical financial statements into a blank Excel sheet?

a) Enter the numbers manually from an annual filing PDF.
b) Link directly to the company's Excel or CSV with the financials, copy and paste as values, and then manipulate and format the data.
c) Link directly to the company's Excel or CSV, but choose the line items to link to as you go down through each statement.
d) Download the financials from Capital IQ or FactSet and use them as-is.

Explanation: The second answer choice is the best one. It's fastest to link to all the text labels and data first and then consolidate, delete, and add lines and apply formatting once you've copied and pasted everything as values. It will take much longer to do this if you copy and paste manually from the filing PDFs or if you selectively link in or copy items from the Excel or CSV file. Downloading the financials from Capital IQ or FactSet is an option as well, but even if you do this, you'll have to spend time changing around the financials to match the presentation you want – and if you don't have the source data, it may be difficult to do that in some cases.

AFM-11-03: Revenue Projections with Scenarios (24:45)

In this lesson, you'll build different operating scenarios for Netflix based on the long-term historical data and its recent Q1 results, and you'll learn how to set up different assumptions for the paid subscribers and average revenue per subscriber.

NOTE TO LESSON: The FY 21 Total Revenue and Streaming Revenue in the "Data" tab are wrong. We fix this later in the module (lesson AFM-11-11), and this error with the data makes almost no difference because it only affects the DVD Revenue in the projections, which is insignificant.

Table of Contents:

- **4:07:** Historical Data and Trends
- **10:47:** Revenue Scenarios
- **20:01:** Checking the Numbers
- **23:03:** Recap and Summary

Quiz Questions:

20:00 Why do we assume much lower paid membership growth rates for Netflix in the projected period vs. the historical period in all scenarios?

a) Poor Q1 results in which the company disclosed the first-ever loss in paid members.
b) Rising competition from streaming services such as Disney+ and HBO Max.
c) Market saturation, especially in regions with higher membership fees.
d) Consensus views that indicate expected revenue growth in the ~10% per year range.
e) All of the above.

Explanation: Everything above is true. While Netflix has grown its paid members by 20-40% historically, this growth slowed sharply in the most recent fiscal year (to under 10%), and its Q1 earnings had evidence of its first-ever *loss* in members. The streaming service market is increasingly saturated with well-funded competitors, and most research analysts expect much slower growth for the company in the future.

AFM-11-04: Income Statement Projections (25:24)

You'll project Netflix's Income Statement, including its Cash Content Spending and Content Amortization, in this lesson, and you'll compare its operating margins in the different cases.

Table of Contents:

- **3:13:** Cost of Revenue
- **14:34:** Operating Expenses
- **16:55:** Linking and Checking the Numbers
- **23:24:** Recap and Summary

Quiz Questions:

14:33 Why do we link Netflix's Cash Content Spending to the total # of paid memberships rather than making it a percentage of revenue?

a) Because even if the revenue per paid member decreases, the company might need to produce and license more content if memberships increase.
b) Because by doing this, we ensure that the margins change at least a bit in the different operating cases.
c) Because all of Netflix's paid members are on month-to-month contracts and tend to cancel and renew based on the currently available content.

d) **All of the above.**

Explanation: Everything above is true. We do want the margins to be slightly different in the different scenarios, and we also want to reflect reality: more paid members mean the company needs to produce or license more content, and since members are on month-to-month contracts, they tend to be quite fickle based on the available content.

AFM-11-05: Balance Sheet and Cash Flow Statement Projections (24:49)

In this lesson, you'll calculate the historical percentages for the key Balance Sheet and Cash Flow Statement drivers, set up the projections, and then link the statements to create a functional 3-statement model.

NOTE TO LESSON: At 20:49 to 20:59, we should also link "Acquisitions" to "Property and Equipment" since it doesn't link to anything else. It makes no difference because it's $0 in each projected year, but it is fixed in the "After" Excel file.

Table of Contents:

- **1:32:** Historical Percentages
- **10:12:** Projections
- **15:19:** Linking the Statements
- **23:15:** Recap and Summary

Quiz Questions:

15:18 Why might you assume lower numbers for Stock-Based Compensation as a % of Revenue in future periods for Netflix?

a) **Because the company's stock price has just fallen by ~70%, so issuances of similar dollar amounts imply much greater dilution.**
b) Because you do not want the SBC add-back on the Cash Flow Statement to boost the company's Free Cash Flow significantly.
c) Because the company's overall operating margins are rising in the projected period.
d) All of the above.

Explanation: The first answer choice is correct. At a much lower stock price, the same dollar amount of SBC would create far more dilution, so Netflix will likely reduce the total amount it issues to employees because of its recent stock price drop. The second answer does not make sense because you can subtract out SBC if you want to avoid this issue, and the third answer

does not make sense because the operating margin alone tells you nothing about the portion of employee expenses paid in cash vs. stock.

AFM-11-06: Financing Scenarios and Equity Value / Enterprise Value Bridge (Debt Comps and Follow-On Equity Comps) (24:35)

You'll set up the Enterprise Value bridge in this lesson and make assumptions for the follow-on equity offering, senior unsecured notes, and convertible bond based on a review of the comparable companies and the comparable debt and follow-on equity offerings.

Table of Contents:

- **1:41:** Eq Val / TEV Bridge
- **4:24:** Capital Raise Assumptions
- **13:47:** Convertible Bond and Senior Note Assumptions
- **23:16:** Recap and Summary

Quiz Questions:

13:46 When might a company have to offer a *higher than normal* discount on its shares in a Follow-On Equity Offering?

a) If the deal size as a percentage of its Market Cap is quite large (above the median for the Follow-On Offering comparables).
b) If the company has recently reported poor financial results or reduced its expected growth.
c) If the company's share price has recently changed by a significant percentage (vs. the average over the past year).
d) All of the above.

Explanation: Everything above is true. If the company has performed poorly recently or guided to lower growth, it's less appealing to equity investors, so it will have to offer a higher discount. The same is true if its share price has recently dropped or increased by a huge percentage – an above-average discount may be required to increase investor interest. Finally, if the company is selling a much higher percentage of itself than in recent comparable deals, it may also need to offer a higher discount due to the greater dilution.

AFM-11-07: Valuation and Diluted Share Count Projections (25:26)

In this lesson, you'll project Netflix's valuation multiples, implied share price, and diluted share count in the different operating scenarios so that you can determine the conversion status of the convertible bond and calculate credit stats and ratios later.

NOTE TO LESSON: In the 15:15 – 15:36 segment here, the formula for the dilution from the convertible bond does not quite handle all the cases correctly. We enter an improved formula in lesson #10.

Table of Contents:

- **2:34:** Projected Valuations
- **9:52:** Share Repurchase, Follow-On, and Convertible
- **16:30:** SBC and Shares from Options
- **23:45:** Recap and Summary

Quiz Questions:

9:51 Which of the following data sources is NOT useful when projecting Netflix's potential EBITDA multiples in different operating cases?

a) The range of EBITDA multiples for the public comps.
b) Netflix's range of EBITDA trading multiples over the past decade.
c) Netflix's projected growth rate and growth patterns vs. its comparable companies.
d) None of the above, i.e., these data sources are ALL useful in this case study.

Explanation: The second answer choice is correct, i.e., it is the LEAST USEFUL data source here. The issue is that Netflix's recent and projected growth rates are far below what they were historically, so the company should trade at much lower multiples. Therefore, it's more relevant to compare it to a set of movie/entertainment public comps and their multiples and growth rates now that the company is more of a media company and less of a high-growth tech company.

AFM-11-08: Debt Schedule, Part 1: Data Gathering (23:35)

You'll start setting up the Debt Schedule in this lesson by gathering and inputting data for the Revolver and existing Senior Note tranches of Debt and consolidating the tranches, where possible, to save time and space.

Table of Contents:

- **3:11:** Revolver and Coupon Rates
- **11:25:** Face Values and Maturities

- **17:19:** Cash Flow Setup
- **21:33:** Recap and Summary

Quiz Questions:

11:24 When building a Debt Schedule, when might you be able to consolidate different tranches of Debt into one?

a) When they have the same or similar coupon rates, maturities, and other terms.
b) When they mature after the end of the projection period in the model.
c) When they mature in the same period, such as the same year in an annual model.
d) All of the above.

Explanation: Everything above is true. These are all factors that might explain why you choose to consolidate certain tranches of Debt. In other words, it's worth modeling separate tranches separately ONLY if they have much different terms, such as different coupon rates, maturities, or mandatory/optional repayments. The more similar the terms are – especially if the Debt matures after the end of the projection period or holding period – the less worthwhile it is to model the tranches separately.

AFM-11-09: Debt Schedule, Part 2: Cash Flow and Revolver (28:08)

In this lesson, you'll set up the trickiest formulas in the Debt Schedule, including the ones for the Revolver with a maximum Draw limit, the convertible bond conversion formulas, and the conditions for additional Senior Note funding and stock repurchases.

Table of Contents:

- **2:03:** Filling in Other Line Items
- **5:35:** Convertible Bond Formulas
- **9:53:** Debt Maturities and Minimum Cash
- **14:21:** Revolver Formulas
- **19:27:** Additional Funding and Stock Repurchases
- **23:27:** Changing Debt Balances
- **25:38:** Recap and Summary

Quiz Questions:

9:53 Which of the following is the BIGGEST difference between the accounting treatments of a convertible bond converting into equity under U.S. GAAP vs. IFRS?

a) **The entire convertible bond moves into Equity when this happens under U.S. GAAP, but under IFRS, part of it is already in Equity.**
b) The IFRS treatment ignores the remaining unamortized issuance fees on the convertible bond.
c) There is a Gain or Loss on Debt Extinguishment recorded on the Income Statement when this happens under IFRS.
d) There is no Amortization of the Convertible Bond Discount under U.S. GAAP (in most cases).

Explanation: The first answer is the best choice. The third and fourth answer choices are true statements, but the third answer is a fairly small difference, and the fourth answer does not describe a difference related to the CONVERSION of the convertible bond into equity. The second answer is wrong because the IFRS treatment does factor in the unamortized issuance fees via the Gain or Loss on Debt Extinguishment. The first answer is the best one because under IFRS, most convertible bonds are split into significant Debt and Equity components, but under U.S. GAAP, most are shown as simple Debt liabilities.

23:26 In which of the following situations would Netflix repurchase shares?

a) Cash Flow Deficit = $200; Revolver Balance = $1,000.
b) **Cash Flow Surplus = $500; Revolver Balance = $400.**
c) Cash Flow Surplus = $800; Revolver Balance = $900.
d) Cash Flow Deficit = $500; Revolver Balance = $0.

Explanation: The second answer is correct. For a stock repurchase to take place, Netflix's cash flow surplus *after* Revolver repayment must be positive. If it has a cash flow deficit, no stock repurchases will ever occur, ruling out the first and fourth answers. The third answer is wrong because with these numbers, the company will repay $800 of its Revolver balance and have nothing left for stock repurchases. Only the second answer gives the company enough cash flow to repurchase stock.

AFM-11-10: Debt Schedule, Part 3: Interest Expense and Financial Statement Links (21:43)

You'll project the Interest Expense and Amortization of Issuance Costs in this lesson, and you'll fix some minor issues with the other schedules and link the Debt & Equity Schedule to Netflix's full financial statement projections.

Table of Contents:

- **1:17:** Interest and Issuance Costs
- **7:56:** Linking the Debt Schedule
- **16:11:** Case Review
- **20:30:** Recap and Summary

Quiz Questions:

16:10 When linking the statements to the Debt & Equity Schedule in the Netflix model, why does the convertible bond's conversion into Equity not appear explicitly on the Cash Flow Statement?

a) Because Netflix follows U.S. GAAP rather than IFRS, and the rules for conversions into Equity are different.
b) Because there's no pre-existing Equity component of the convertible bond before the conversion takes place.
c) Because there's no net change in Cash from the reclassification of the convertible bond into Equity.
d) All of the above.

Explanation: The third answer choice is correct. Generally, items that do not make a net impact on the company's Cash balance do not show up explicitly on the Cash Flow Statement, so the reclassification of convertible bonds from Debt into Equity qualifies as such a line item or change. While the first two answer choices contain partially correct statements, these statements do not explain why the conversion into Equity is not shown on the CFS; also, the rules for conversions under IFRS are only slightly different, so the first answer choice is a poor potential explanation here.

AFM-11-11: WACC Analysis Before and After Debt/Equity/Convertible Bond Issuance (29:36)

In this lesson, you'll calculate WACC before and after the company's capital issuance and build a sensitivity table that shows how WACC changes based on the Debt, Equity, and Convertible Bond percentage split in the $4 billion capital issuance.

Table of Contents:

- **2:23:** Model Setup and Changes
- **8:26:** WACC Assumptions
- **14:30:** WACC Calculations
- **21:25:** WACC Sensitivities

- **28:10:** Recap and Summary

Quiz Questions:

14:29 When a convertible bond switches from being treated as Debt to Equity, what should happen to a company's overall Pre-Tax Cost of Debt?

a) It should increase because more Debt means a higher Cost of Debt.
b) It should decrease because convertible bonds have lower coupon rates than traditional Debt.
c) It depends on the convertible bond's YTM when treated as traditional Debt vs. the company's current weighted average YTM.

Explanation: The last answer choice is correct. In this specific example with Netflix, its Pre-Tax Cost of Debt goes up when the convertible bond is counted as 100% traditional Debt – but that's only because its Yield to Maturity exceeds the weighted YTM of the existing issuances. It could go either way, depending on whether this is a new or existing issuance and the relative YTMs. If interest rates have been rising, treating potential or newer issuances as Debt will tend to increase the Cost of Debt, and the opposite is true if interest rates have been falling since the time of most of the company's existing issuances.

AFM-11-12: Credit Stats and Ratios (25:27)

You'll calculate the key credit stats and ratios in this lesson, such as the Leverage Ratio, Interest Coverage Ratio, Debt Service Coverage Ratio (DSCR), Fixed Charge Coverage Ratio (FCCR), and Debt / Total Capital Ratio, and you'll see how they change in the different operating cases.

NOTE TO LESSON: We start calculating the Fixed Charge Coverage Ratio (FCCR) between 15:15 and 15:35 without explaining what "Fixed Charges" mean. Please see the Recap and Summary for the explanation or refer to the written notes in the Excel files.

Table of Contents:

- **2:16:** Step 1: Link in the Numbers
- **8:45:** Step 2: Calculations and Meanings
- **19:03:** Step 3: Outcomes in Different Scenarios
- **23:38:** Recap and Summary

Quiz Questions:

19:02 Why is it important to look at FCF-based credit stats and ratios such as the Debt Service Coverage Ratio for Netflix?

a) Because the company's main constraint is the Interest Expense, which EBITDA-based metrics ignore.
b) Because FCF is the only metric that deducts *all* of the company's Fixed Charges (interest, debt principal repayments, and lease payments).
c) Because Netflix has a significant amount of mandatory principal repayments on its loans, and it will use its FCF to pay for them.
d) Because unlike EBITDA, FCF deducts the company's net cash spent on content (EBITDA only deducts amortized content spending).

Explanation: The last answer choice is correct. The key issue for Netflix is that only FCF deducts the Cash Content Spending, net of Content Amortization – EBITDA deducts only the Content Amortization, which is deceptive since it's quite different from the actual cash spending. The first answer is wrong because the Interest Expense is not the key constraint, as the EBITDA Cushions are quite high with that metric, and the third choice is wrong because the Senior Unsecured Notes have no mandatory principal repayments. The second choice is a true statement, but it's irrelevant because when you use FCF in a metric like the DSCR, you add back the Fixed Charges first anyway.

AFM-11-13: Model Tweaks and How to Pick the Best Financing Mix (13:59)

In this lesson, you'll tweak the Downside Case of the model and experiment with different numbers for the Debt and Equity percentages to determine the best financing mix for Netflix's $4 billion capital raise.

Table of Contents:

- **0:31:** Known Facts
- **2:36:** Downside Case Changes
- **4:17:** Financing Mix for the Capital Raise
- **12:57:** Recap and Summary

Quiz Questions:

[No quiz questions.]

AFM-11-14: Decision-Making Process, Case Study Presentation, and Memo (14:06)

You'll learn about the full decision-making process for financing decisions in this lesson, and you'll see how it's implemented in the case study presentation and credit memo that recommend a specific mix of debt and equity for Netflix.

Table of Contents:

- **1:19:** Decision-Making Process
- **7:10:** Case Presentation and Answers
- **12:47:** Written Memo

Quiz Questions:

12:47 Which of the following conditions would have to be true for a convertible bond to make more sense than traditional debt or equity for Netflix?

a) Cash interest expense would have to be a key constraint.
b) The company would have to be in a higher growth category or have more stock-price appreciation potential.
c) Metrics such as Debt / EBITDA and Debt / Total Capital would have to be fine or not relevant for the company.
d) All of the above.

Explanation: Everything above is true. Convertibles are not "cheaper" than traditional debt or equity since they're counted as either one or the other in the WACC calculation; their main advantage is a lower cash interest expense than traditional debt. Also, not all types of companies can issue convertible bonds since they tend to be most common among high-growth, speculative companies such as biotech startups and technology companies in growth mode. Investors often view them as "hedged equity," so they tend not to buy convertible issuances from lower-growth/mature companies. Finally, financial maintenance covenants are rare/non-existent on convertible bonds, but if the company targets specific numbers based on its overall credit profile, a convertible bond counted as debt would still affect numbers such as Debt / EBITDA, which explains why the last answer choice is also correct.

AFM-11-15: Introduction to Bond Analysis and Bond Indentures (20:07)

In this lesson, you'll analyze a bond indenture for one of Netflix's Unsecured Senior Notes and fill in the key terms, such as the market price, the coupon rate, the interest payment dates, and the limitations on indebtedness when issuing new notes.

Table of Contents:

- **2:07:** Part 1: What Makes Bond Analysis Trickier Than It Seems
- **6:39:** Part 2: How to Use Bond Analysis in Investment Banking
- **10:00:** Part 3: How to Complete a "Bond Indenture Analysis"
- **18:29:** Recap and Summary

Quiz Questions:

18:28 Why is a "limitation on indebtedness" in a bond indenture not necessarily the same thing as a financial maintenance covenant that restricts a company's Debt / EBITDA?

a) The financial maintenance covenant normally requires compliance with a number that is measured repeatedly (e.g., each quarter).
b) Limitations on indebtedness might refer to the company's credit stats only when it is issuing new debt under similar terms.
c) The criteria might be slightly different, such as a maximum debt amount OR a maximum Debt / EBITDA.
d) All of the above.

Explanation: Everything above is true. Limitations on indebtedness are not normally considered "covenants" in quite the same way as traditional maintenance covenants because of these issues: they apply when a company is issuing *new* notes, but they don't necessarily need to be maintained at all times. And unlike incurrence covenants, they do not restrict the company from taking specific actions (except for issuing new debt).

AFM-11-16: Bond Yields, Pricing, and Key Relationships (29:52)

You'll learn how to calculate bond prices and different types of yields in this lesson, and you'll learn about the key relationships between coupon rates, bond prices, the discount rate, the yield to maturity, the time to maturity, and other factors.

Table of Contents:

- **1:25:** Part 1: Assumptions and Excel Setup
- **9:32:** Part 2: The Yield and Yield to Maturity (YTM)
- **16:57:** Part 3: Bond Price Calculations
- **21:16:** Part 4: Key Bond Relationships and Graphs
- **28:09:** Recap and Summary

Quiz Questions:

16:56 Why might the Excel YIELD function, the XIRR function, and the quick approximation method for the Yield to Maturity each produce slightly different results?

a) This might happen due to slight timing differences, such as 365 days in the year vs. the 360-day convention.
b) The treatment of accrued interest or interest paid in stub periods in between coupons might be different.
c) The dates might be less precise in the "quick approximation method" (e.g., simple whole numbers for the years rather than decimals).
d) All of the above.

Explanation: Everything above is true. Dates are treated slightly differently by all these YTM calculation methods, which explains most of the minor discrepancies. Accrued interest should be handled roughly the same way, but sometimes the internal Excel functions and manual IRR/XIRR functions do not line up, so that can also create differences.

AFM-11-17: Bond Duration, Convexity, and Key Relationships (33:12)

In this lesson, you'll learn what a bond's Duration and Convexity mean, how to calculate them, and how to use these properties in real life when advising clients on financing and refinancing decisions and making investment recommendations.

Table of Contents:

- **1:21:** Part 1: Duration and Convexity Overview
- **4:08:** Part 2: Bond Duration: Intuition and Calculations
- **19:24:** Part 3: Bond Convexity: Intuition and Calculations
- **28:28:** Part 4: How to Use These Concepts in Investment Banking
- **30:54:** Recap and Summary

Quiz Questions:

19:23 Suppose that Bond A has a Duration of 6.0 and Bond B has a Duration of 10.0. Which of the following is a VALID interpretation of these numbers?

a) Investors in Bond A are exposed to more interest-rate risk due to the shorter Duration.
b) When the Discount Rate changes by 1%, Bond B's Price should change by 10%.
c) Bond A likely has a higher Coupon Rate than Bond B due to its shorter Duration.
d) All of the above.

Explanation: The second answer choice is correct. This is the only correct interpretation of Duration here; a Duration of 10.0 means the bond's Price will change by 10% when the Discount Rate changes by 1%, since Duration is the first derivative of Price with respect to the Discount Rate or YTM. The first answer is false because the interest-rate risk is lower when the Duration is shorter, and the third answer is false because you cannot infer anything about the Coupon Rate based on the Duration. The time to maturity, market price, or other factors might also explain the difference in Duration.

AFM-11-18: Accrued Interest, Clean Prices, and Dirty Prices (14:34)

You'll learn about accrued interest and clean and dirty prices for bonds in this lesson, and you'll see examples of how these factors affect the formulas and output for the bond's price, yield to maturity, duration, and convexity.

Table of Contents:

- **0:51:** Part 1: "Clean" vs. "Dirty" Bond Prices
- **2:17:** Part 2: Excel Setup and the COUPPCD Function
- **7:55:** Part 3: How the "Dirty Price" Changes Everything
- **12:43:** Recap and Summary

Quiz Questions:

[No quiz questions.]

AFM-11-19: Call and Put Options, the Yield to Call (YTC), Yield to Put (YTP), and Yield to Worst (YTW) (22:03)

In this lesson, you'll learn about call and put options on bonds and how they affect the Coupon Rate and other terms; you'll use them to calculate the Yield to Call and Yield to Put on different dates and determine the Yield to Worst (YTW).

Table of Contents:

- **2:00:** Part 1: Why Call and Put Options on Bonds Exist
- **4:43:** Part 2: Call Options, Yield to Call (YTC), and Yield to Worst (YTW)
- **15:22:** Part 3: Put Options and Yield to Put (YTP)
- **20:01:** Recap and Summary

Quiz Questions:

15:21 Suppose that a bond's YTW is 5%, and its YTM is 6%. Which of the following can you conclude based on this data?

a) **The Discount Rate (i.e., prevailing yields on similar bonds) is currently less than or equal to the bond's coupon rate.**
b) The Discount Rate is currently greater than the bond's coupon rate.
c) The Discount Rate is currently equal to the bond's coupon rate.
d) It's unlikely that the bond has significant call premiums (call redemption prices) attached.

Explanation: The first answer is correct. If the YTW is less than the YTM, all you can conclude is that the Discount Rate is less than or equal to the bond's Coupon Rate (based on the pattern of call premiums and the relative difference). This is because the bond trades at a premium or par value when this is true, so if the company repays the bond early, the investors can lose some yield if the premium paid in the beginning exceeds the call redemption earned back upon early repayment. The second and third answers misstate this relationship (and the third one is true only if the YTW equals the YTM). The fourth answer may be true, but it's impossible to say from just this data, so it's not the best choice.

AFM-11-20: Make-Whole Analysis and Make-Whole Prices (18:57)

You'll learn about Make-Whole Provisions and Make-Whole Prices on bonds in this lesson, and you'll understand why they heavily discourage companies from repaying their bonds very early on (e.g., Years 1 – 3 for a 10-year bond).

Table of Contents:

- **2:22:** Part 1: The Idea Behind Make-Whole Provisions
- **6:51:** Part 2: Make-Whole Price Calculations
- **14:48:** Part 3: What the Make-Whole Price Means for Netflix
- **17:19:** Recap and Summary

Quiz Questions:

17:18 Why does a Make-Whole Provision for early repayment of a bond ensure that it's usually NOT a great idea for the company to repay its bond early?

a) If interest rates have fallen, the penalty fee for early repayment will be much higher.

b) If interest rates have risen, the penalty fee will be lower, but the company will pay a higher Coupon Rate on a new issuance.
c) Penalty fees for early repayment almost always decrease further into the bond's tenor.
d) All of the above.

Explanation: Everything above is true. Make-Whole Provisions heavily discourage early repayment because regardless of whether interest rates have fallen or risen since the bond's initial issuance, the company loses out: either the penalty fee for early repayment is higher, or the penalty fee is lower, but the company will have to pay a higher Coupon Rate on the new issuance. *After* the Make-Whole Provision switches to normal call redemption prices that are fixed percentages of par value – if that happens – then it makes more sense for a company to consider early repayment if interest rates have fallen significantly or its credit profile has improved.

AFM-11-21: Bond Refinancing Recommendation for Netflix (27:58)

In this lesson, you'll put together all the previous lessons to make a refinancing recommendation to Netflix, and even if you recommend against the refinancing, you'll still suggest possible terms for a new bond issuance based on investors' targeted YTM, Duration, and Convexity, and the company's desire for reduced interest expense.

Table of Contents:

- **1:24:** Part 1: Should Netflix Refinance?
- **4:50:** Part 2: Running the Numbers
- **5:28:** Cash Flow Setup
- **13:15:** Sensitivities
- **16:05:** Duration and Convexity for New Issuance
- **20:35:** Part 3: Likely Terms If Netflix Does Refinance
- **26:34:** Recap and Summary

Quiz Questions:

26:33 Suppose that a company wants to refinance and offer investors a similar YTM on a new bond by issuing it at a lower Coupon Rate but also including a substantial Original Issue Discount (OID). What is the major disadvantage of this plan?

a) The Duration and Convexity will be much lower.
b) The company is likely to "lose money" because it repays the first bond at par value and then gets less than par value for the new bond.

c) The company's credit rating will likely fall, which will result in a slightly lower YTM.
d) All of the above.

Explanation: The second answer is the best choice for the reasons stated: if a company refinances a $1 billion bond and offers a 5% OID on the new bond, effectively, it loses $50 million since it repays a bond for $1 billion but gets only $950 million in proceeds from the new bond. The Duration and Convexity will be slightly lower due to the OID, but the difference is so small that this is not a good answer. The third answer does not make sense because, if anything, the company's credit rating should be higher if it issues the bond at a lower Coupon Rate and improves its EBITDA / Interest ratio (and others).

AFM-11-22: Mini-Case Study: How to Make a Refinancing Recommendation to a Distressed Company [OPTIONAL] (41:06)

You'll complete an optional case study in this lesson and analyze a distressed company's current bond issuance and prospective new bond issuance to make a recommendation about how the company might change the terms to appeal to investors more effectively.

Table of Contents:

- **3:15:** Part 1: Case Overview / Summary
- **6:17:** Part 2: Bond Assumptions
- **12:35:** Part 3: Capital Structure Projections
- **20:23:** Part 4: Cash Flows and Yield to Exit
- **28:30:** Part 5: Alternate Bond Terms
- **39:04:** Recap and Summary

Quiz Questions:

39:03 Suppose that you recommend a higher Coupon Rate in a new bond issuance for a distressed company that already has a relatively low Cash balance. Which of the following bond terms could be constrained or affected by the Cash balance that changes based on this new rate?

a) Original Issue Discount (OID).
b) Redemption Prices for early repayments.
c) Equity Grants.
d) All of the above.

Explanation: Everything above is true. A higher Coupon Rate will reduce the company's cash balance, which means it won't be able to afford OID as easily – or at all – and that it will have

less available to pay for the penalty fees on early repayments ("Redemption Prices"). Also, a lower cash balance will affect the Equity Grants as well – though by much less than the other items on this list – because less cash means the company's Implied Equity Value will be lower, at least if you forecast its Enterprise Value and back into Equity Value based on that.

AFM-11-23: Convertible Bond Valuation and Payoff Diagrams (36:49)

In this lesson, you'll learn how to value a convertible bond by splitting it into its Liability and Equity components and using Black-Scholes to estimate the market value of the call option on the bond; you'll then create sensitivities and a payoff diagram for the bond based on this analysis.

Table of Contents:

- **3:49:** Part 1: Liability Component
- **8:22:** Part 2: Call Option Component
- **16:25:** Excel Implementation of Black-Scholes
- **23:31:** Sensitivity Tables
- **27:45:** Issues with Black-Scholes
- **29:35:** Part 3: Convertible Bond Payoff Diagram
- **33:15:** Part 4: Conclusions About Netflix's Issuance
- **35:12:** Recap and Summary

Quiz Questions:

16:24 Which of the following assumptions makes the greatest impact on a call option's value under Black-Scholes?

a) The Risk-Free Rate.
b) The time to expiration or maturity.
c) The stock-price volatility.
d) The company's current stock price.

Explanation: In general, the volatility is the most important driver of a call option's value because it affects both terms of the formula (the stock price distribution and the exercise price distribution) and both the numerator and denominator of "d1" in the Black-Scholes formula. The Risk-Free Rate and time to expiration both make less of an impact. The current stock price also makes a difference, but mostly *in relation* to the exercise price – the closer they are, the more valuable the option. So, the fourth answer choice is not the best one here because it could be stated more accurately.

33:14 Suppose that a company's current stock price is $150, and the conversion price of its convertible bond is $200. The convertible bond's face value is $1 billion. Based on ONLY this information, which of the following is a reasonable inference?

a) Currently, the bond's market price is below its payoff value.
b) The bond's market price will equal its payoff value when the stock price rises to $200.
c) When the stock price is above $200, the bond's market price will exceed $1 billion.
d) None of the above.

Explanation: The third answer choice is the best one. When the stock price exceeds the conversion price, the convertible bond's market price *must* be above its face value because the bond also offers downside protection to investors, unlike normal shares. We have no idea whether the first inference is true because it depends on the company's volatility and the other inputs into Black-Scholes, and the second inference is likely false because the market price usually equals the payoff value well before the stock price reaches the conversion price.

AFM-11-24: Convertible Bond Analysis: Blended Cost, Other Metrics, and Capped Call Transactions (31:15)

You'll finish the convertible bond analysis in this lesson by calculating its blended cost based on the cost of the embedded call options and calculating related metrics, such as the payback period and the dilution at different share prices if the company sets up a "capped call" or note hedge transaction along with the bond.

Table of Contents:

- **2:00:** Part 1: The Blended Cost of Convertible Bonds
- **12:48:** Part 2: Additional Calculations and Metrics
- **19:45:** Part 3: Capped Call Transactions and "Note Hedges"
- **29:36:** Recap and Summary

Quiz Questions:

12:47 If a convertible bond's conversion price increases, and nothing else changes, will the Blended Cost increase or decrease?

a) It depends on whether the conversion price is currently above or below the stock price.
b) It will decrease because the convertible bond is now less like Equity.
c) It will increase because the call option component is now more valuable.

d) It will stay about the same because the lower probability of conversion is offset by the higher Cost of Debt.

Explanation: The second answer choice is correct. If the convertible bond is less likely to convert into Equity, the Blended Cost decreases because the Cost of Equity is higher than the Cost of Debt, and the Cost of the Call Option is higher than even the Cost of Equity. And a higher conversion price – regardless of the current stock price relative to the conversion price – means a lower probability of conversion into Equity. The last answer makes no sense because the Cost of Debt will not increase here, and the third option also does not make sense because the call option becomes less valuable when the conversion price or exercise price is higher.